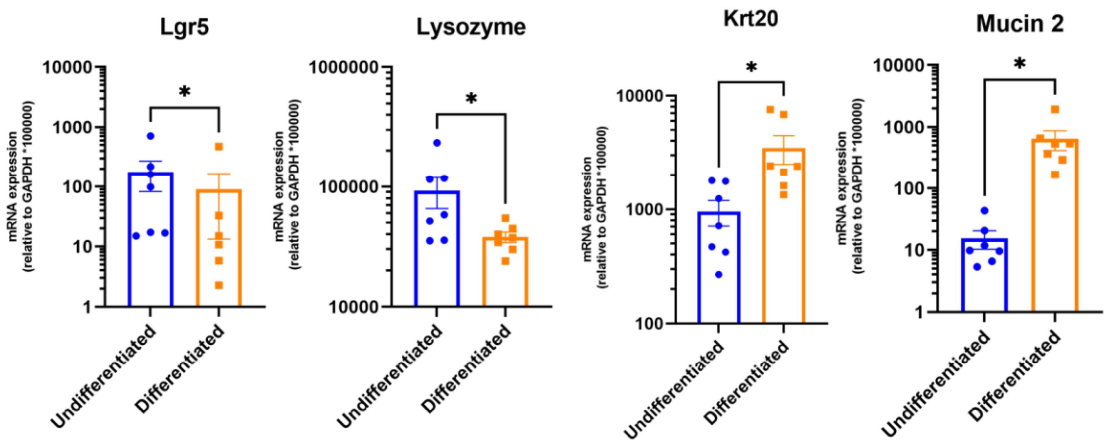


Figure 3.9: RT–qPCR analysis of differentiated and undifferentiated appendix-derived gut organoids. Gene expression of lysozyme (Paneth cells), KRT20 (enterocytes), MUC2 (goblet cells), Lgr5 (intestinal stem cells) was quantified in undifferentiated and differentiated organoids. Data presented as mean ± SD. Statistical analysis was conducted using the Wilcoxon signed rank test or paired t test. *, P<0.05.

3.5 Droplet-Engineered Organoids (DEO)

3.5.1 Droplet-Engineered Organoids (DEO) Generation from Appendix-derived Organoids

Droplet-Engineered Organoids (DEOs) were produced using a combination of microdroplet templating and automated bioprinting, allowing scalable, high-throughput generation of organoids while requiring minimal numbers of input cells [141]. A typical droplet-engineered organoid (DEO) consisted of a single microdroplet with a diameter of approximately 500 µm and a volume of ~0.1 µL. By comparison, conventional organoid culture using the manual pipetting dome method requires deposition of a 50 µL Matrigel dome per well for the establishment of appendix-derived organoids [143]. Thus, the droplet-engineering strategy increases experimental throughput by more than two orders of magnitude while substantially reducing variability associated with manual handling. Moreover, the distinct geometric constraints of the DEO system are likely to influence biological behavior [152] [153]. In DEOs, cells are confined within a limited radius from the droplet center, which is expected to promote cell–cell interactions.

To examine the impact of initial cell density on organoid growth within DEOs, crypts derived from well-established organoids cultured in conventional 50 µL domes were dissociated and quantified. The crypts were then allocated into low-, middle-, and high-density groups prior to encapsulation in DEOs (Figure 3.10A). Organoid growth was monitored longitudinally by brightfield imaging from day 0 to day 16. In the low-density condition, organoid growth was delayed, with only small cystic structures becoming evident around days 8–9. Although differentiation appeared to proceed appropriately, the slow expansion and increased inter-organoid spacing likely limited effective cell–cell interactions. In contrast, organoids seeded at middle density reached sizes comparable to those observed in standard cultures by days 4–5 and continued to expand between days 7 and 10 while largely maintaining a cystic morphology. From days 11 to 16, early signs of differentiation became apparent, and closely positioned organoids began to undergo fusion events, which were clearly visible on day 16 under higher magnification. Organoids in the high-density group exhibited more rapid growth, achieving moderate size within 2–3 days. The close proximity of cells facilitated extensive cell–cell interactions, and organoid fusion was observed as early as days 7–9. However, these cultures also displayed earlier thickening of the epithelial wall, indicative of premature differentiation and the need for earlier passaging. Fusion events were likewise more frequent and occurred at earlier time points compared with the middle-density condition. These findings were reproduced in an independent experiment (Figure 3.10B), which similarly demonstrated slower growth and delayed differentiation at low density, alongside enhanced cell–cell interactions and increased fusion events at intermediate and high densities. Based on these observations, middle- to high-density DEOs were selected for subsequent experiments.

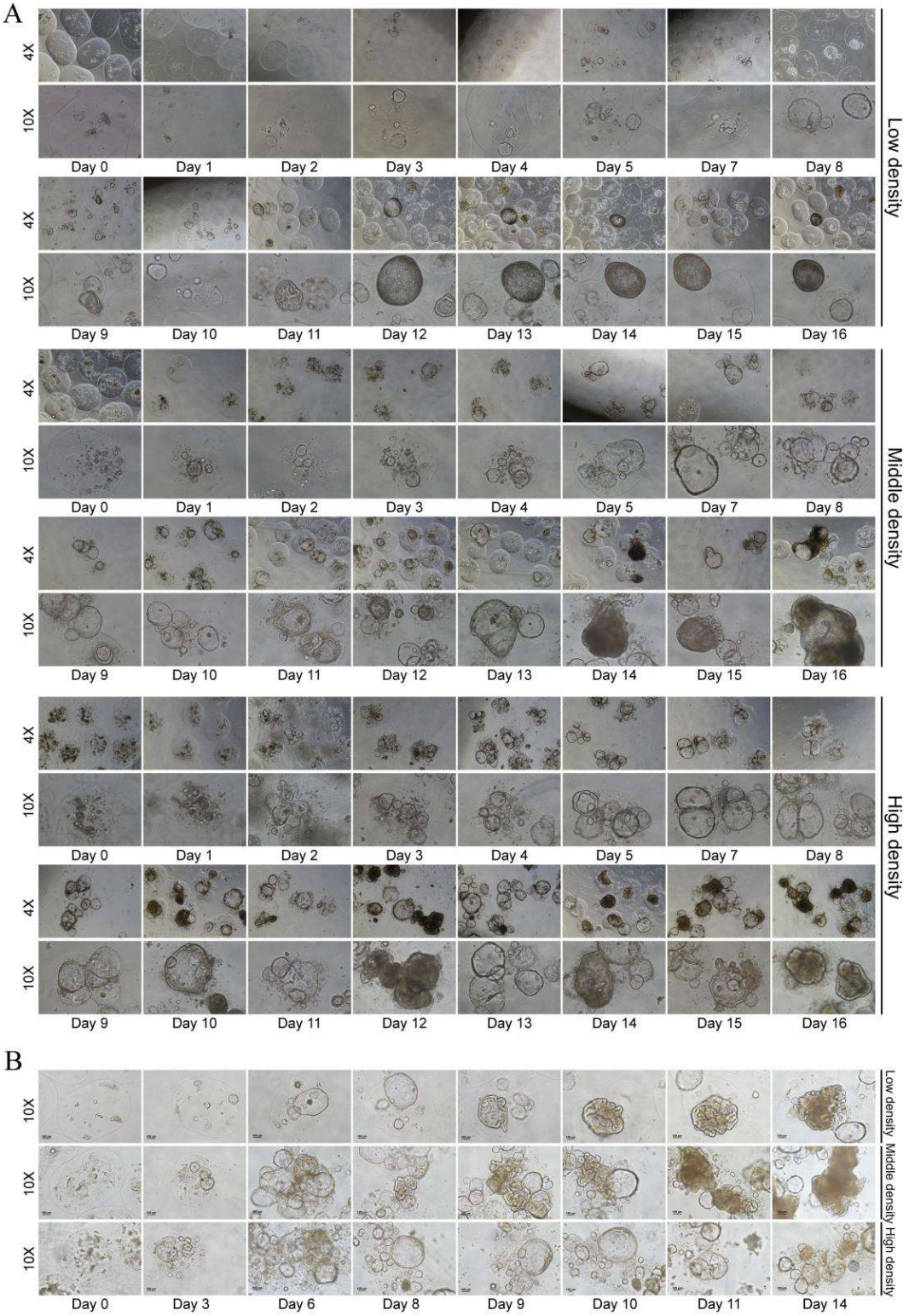

A
4X
10X
Day 0
Day 1
Day 2
Day 3
Day 4
Day 5
Day 7
Day 8
Day 9
Day 10
Day 11
Day 12
Day 13
Day 14
Day 15
Day 16
Low density
Middle density
High density
B
Day 0
Day 3
Day 6
Day 8
Day 9
Day 10
Day 11
Day 14
Low density
Middle density
High density

Figure 3.10 Effects of initial cell density on the growth of Droplet-Engineered Organoids (DEOs). (A) Schematic overview of DEO generation using low-, middle-, and high-density crypt suspensions derived from appendix organoids. Representative brightfield images show longitudinal growth of DEOs from day 0 to day 16. (B) Representative images from an independent experiment confirming density-dependent differences in organoid growth kinetics, differentiation-associated morphological changes, and fusion behavior. Images were acquired using a 4× and 10× objective.

3.5.2 Droplet-Engineered Organoids (DEO) Incorporation with Alginate Sodium

Matrigel provides a permissive extracellular matrix that supports cell adhesion, outward migration, and matrix remodeling due to its high degradability [154]. Previous studies have demonstrated that volumetric compression using alginate can modulate brain organoid development without impairing viability and may even promote tissue growth [144]. Importantly, such compressive environments have been shown to influence organoid morphogenesis rather than inhibit growth. Based on these observations, alginate was incorporated into DEOs to investigate whether organoid growth dynamics and fusion behavior could be regulated through mechanical modulation. Organoids at middle to high cell density were resuspended in Matrigel supplemented with sodium alginate at concentrations of 0.2% or 0.5% (w/v), resulting in final alginate concentrations of 0.02% (low alginate) and 0.05% (high alginate). DEOs generated in Matrigel without alginate served as controls.

Longitudinal brightfield imaging from day 0 to day 13 revealed that alginate-containing DEOs exhibited delayed morphological signs of differentiation compared with control organoids. While control DEOs developed thicker epithelial walls indicative of differentiation as early as days 8–9, alginate-based DEOs maintained cystic structures until approximately days 10–11, with pronounced differences evident by day 13 at higher magnification. Although early growth appeared modestly slower in alginate-containing DEOs during the first five days of culture, overall organoid viability and expansion were preserved. No overt morphological differences were observed between low- and high-alginate conditions based on brightfield microscopy (Figure

3.11).

To assess organoid fusion dynamics, representative 10× brightfield images from day 7 and day 13 were compared across the three experimental conditions. On day 7, evident fusion events were observed in the control group, whereas fusion was also detectable in alginate-containing groups but involved smaller fused structures. In the control condition, a higher proportion of organoids appeared to extend beyond the Matrigel droplets, while organoids cultured with alginate predominantly remained confined within the droplet matrix. By day 13, the overall extent of fusion in the control group showed no marked progression compared with day 7. In contrast, alginate-based cultures displayed modestly increased proximity and cell–cell interactions relative to earlier time points, suggesting a gradual enhancement of fusion-related contacts (Figure 3.12).

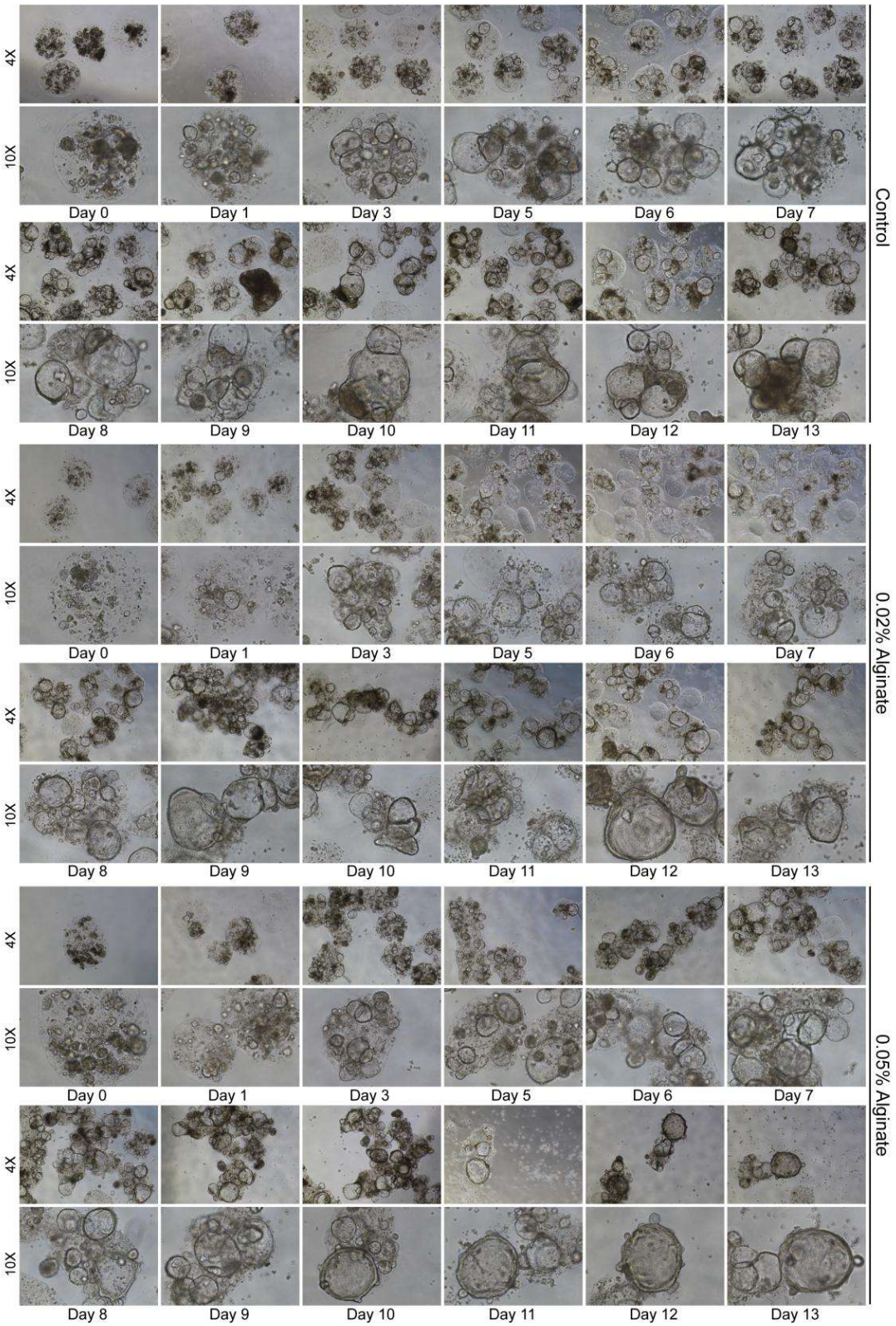

4X
10X
Day 0
Day 1
Day 3
Day 5
Day 6
Day 7
Day 8
Day 9
Day 10
Day 11
Day 12
Day 13
Control
0.02% Alginate
0.05% Alginate

Figure 3.11 Alginate incorporation of droplet-engineered organoids. Brightfield microscopy images of DEOs generated at high cell density in Matrigel alone (control) or Matrigel supplemented with sodium alginate at final concentrations of 0.02% (low alginate) or 0.05% (high alginate). Representative images were captured longitudinally from day 0 to day 13. Images were acquired using 4× and 10× objective lenses, as indicated.

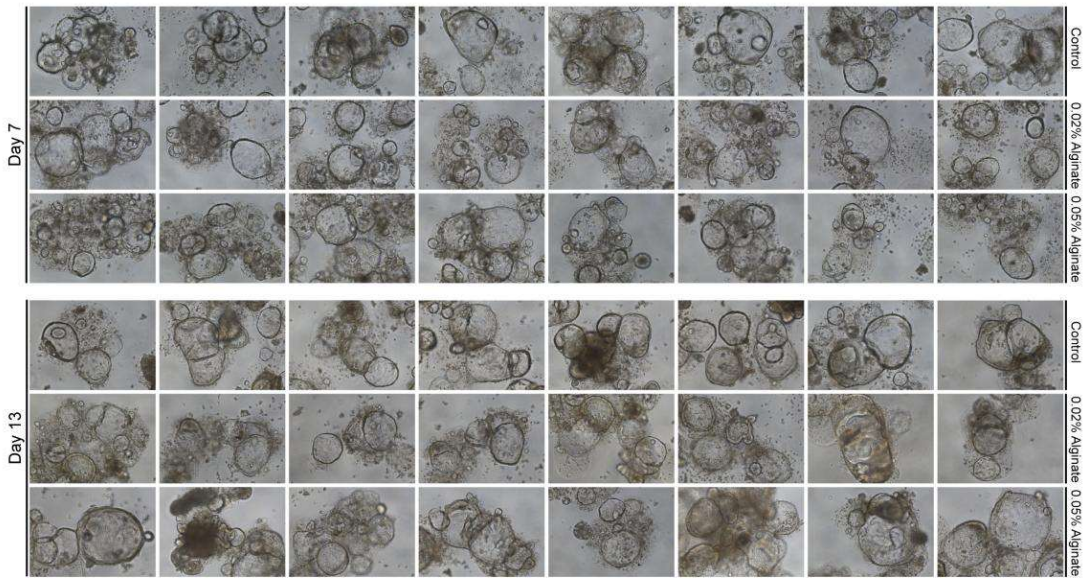


Figure 3.12 Effects of alginate incorporation in droplet-engineered organoids. Representative brightfield images (10× objective) showing organoid fusion behavior in control (Matrigel-only) and alginate-containing DEOs at day 7 and day 13 of culture.

3.5.3 RNA-seq Analysis of Alginate-Incorporated Droplet-Engineered Organoids (DEO)

To investigate transcriptomic differences between control DEOs and alginate-containing DEOs, RNA sequencing was performed on samples from the no-alginate, low-alginate, and high-alginate groups. A Venn diagram summarizes the overlap of differentially expressed genes (DEGs) identified in the no_alg versus low_alg group and no_alg versus high_alg group comparisons (Figure 3.13A). In the

no_alg/low_alg comparison, 676 DEGs were detected, including 454 upregulated and 222 downregulated genes. The no_alg/high_alg comparison revealed 1,354 DEGs, consisting of 342 upregulated and 1,012 downregulated genes (Figure 3.13B).

Sample reproducibility and overall transcriptomic similarity were evaluated using Pearson correlation analysis. All samples showed strong correlations, with coefficients ≥ 0.977, indicating high consistency across replicates (Figure 3.13C). Heatmap visualization of DEGs highlighted pathways associated with organoid fusion and cell–matrix interactions (Figure 3.13D). Within the KEGG cell adhesion molecules pathway (KEGG 04514), CLDN2 expression was markedly increased in alginate-containing groups. Genes associated with focal adhesion signaling (KEGG 04510), including FLNA, FN1, and ITGA8, showed a progressive increase in expression from the no-alginate group to the low-alginate group and further to the high-alginate group. In addition, genes involved in tight junction (KEGG 04540) and gap junction (KEGG 04520) pathways, such as CLDN2, CLDN10, CDK4, and TUBA1A, were upregulated in alginate-treated DEOs.

Differential expression significance was further visualized using a volcano plot, in which the top 20 DEGs were highlighted based on a threshold of Q value ≤0.05 and Log2 fold change (|log2FC|) ≥ 1 (Figure 3.13 E). Functional enrichment analyses using KEGG and databases indicated that DEGs were predominantly associated with pathways related to DNA replication and cell cycle in low_alg/no_alg and high_alg/no_alg comparisons (Figure 3.13 F). Gene Ontology biological process analysis similarly indicated enrichment in cell division, DNA replication, and cell cycle–associated processes in low_alg/no_alg and high_alg/no_alg comparisons (Figure 3.13 G). To validate differentiation status inferred from brightfield imaging, the expression of six representative differentiation-related marker genes was examined (Figure 3.13H). Lgr5 and MKI67 were downregulated in the no-alginate control group compared with both alginate groups, with the high-alginate group maintaining higher expression levels than the low-alginate group. Conversely, expression of epithelial and differentiation markers, including EpCAM, TFF3, KRT20, and FABP2, was reduced following alginate incorporation in a concentration dependent manner. Together, these transcriptomic findings corroborate the morphological observations and demonstrate that alginate incorporation delays or attenuates DEOs differentiation in a dose-dependent manner.

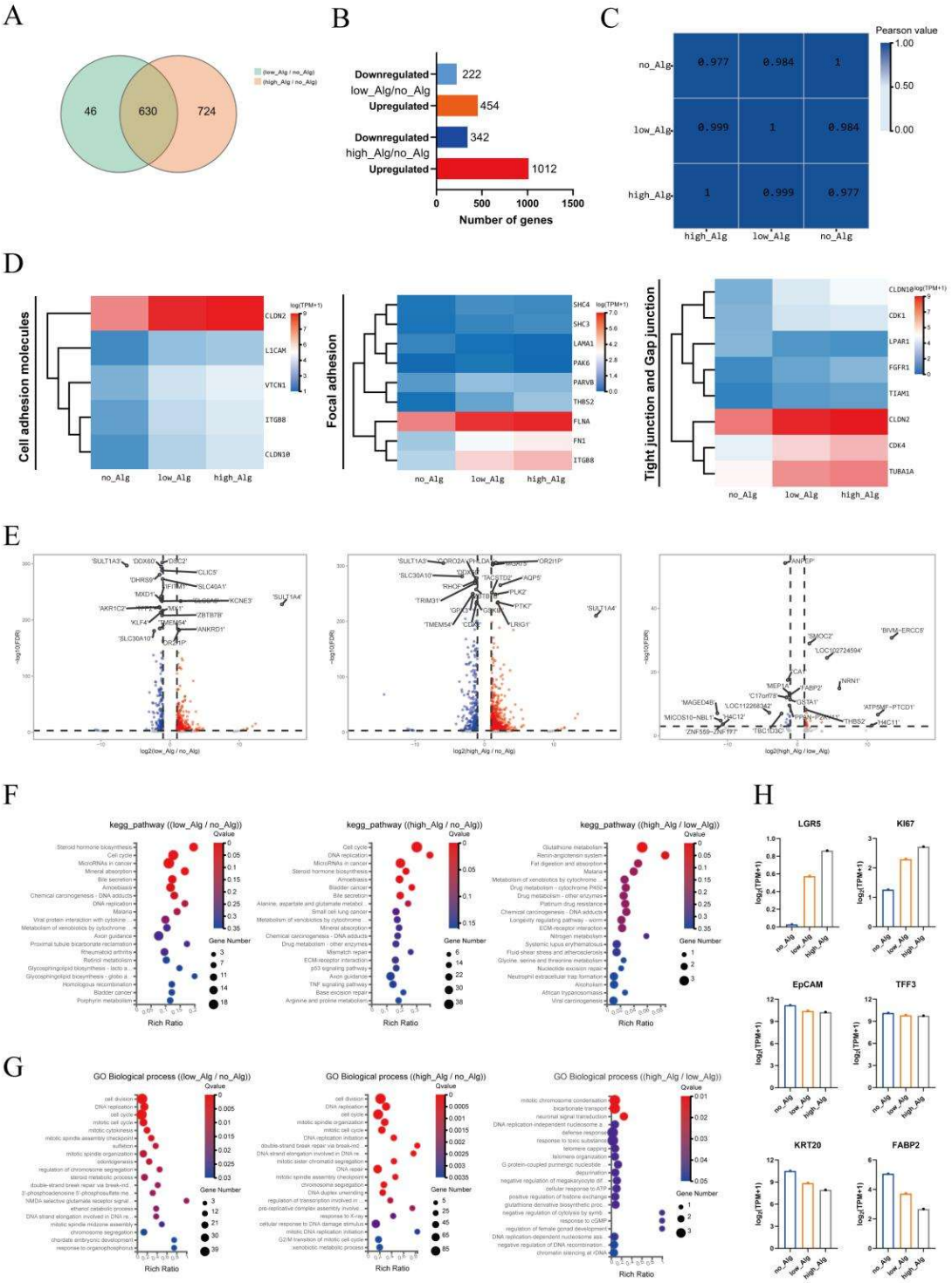

A
46
630
724
B
Downregulated
low_Alg/no_Alg
222
Upregulated
454
Downregulated
high_Alg/no_Alg
342
Upregulated
1012
0
500
1000
1500
Number of genes
C
Pearson value
1.00
0.50
0.00
no_Alg
low_Alg
high_Alg
0.977
0.984
1
0.999
1
0.984
1
0.999
0.977
D
Cell adhesion molecules
CLDN2
L1CAM
VTCN1
ITGB8
CLDN10
Focal adhesion
SHC4
SHC3
LAMA1
PAK6
PARVB
THBS2
FLNA
FN1
ITGB8
Tight junction and Gap junction
CLDN10
CDK1
LPAR1
FGFR1
TIAM1
CLDN2
CDK4
TUBA1A
E
F
kegg_pathway ((low_Alg / no_Alg))
kegg_pathway ((high_Alg / no_Alg))
kegg_pathway ((high_Alg / low_Alg))
Qvalue
Gene Number
Rich Ratio
G
GO Biological process ((low_Alg / no_Alg))
GO Biological process ((high_Alg / no_Alg))
GO Biological process ((high_Alg / low_Alg))
H
LGR5
KI67
EpCAM
TFF3
KRT20
FABP2

Figure 3.13 Transcriptomic analysis of DEOs cultured with or without alginate. (A) Venn diagram showing the overlap of differentially expressed genes (DEGs) between no-alginate/low-alginate and no-alginate/high-alginate DEO comparisons. (B) Number of upregulated and downregulated DEGs identified in no-alginate/low-alginate and no-alginate/high-alginate comparisons. (C) Pearson correlation heatmap among RNA-Seq samples. (D) Heatmap of DEGs associated with pathways including cell adhesion molecules pathway (KEGG 04514), focal adhesion (KEGG 04510), tight junction (KEGG 04540) and gap junction (KEGG 04520) pathways. (E) Volcano plot highlighting the top 20 DEGs in no-alginate/high-alginate comparison based on Q value ≤ 0.05 and |log2 fold change| ≥ 1. (F) KEGG and (G)Gene Ontology (Biological Process) enrichment analyses of DEGs. (H) Expression analysis of representative differentiation-related marker genes, including Lgr5, KI67, EpCAM, TFF3, KRT20, and FABP2. TPM, transcripts per million.

3.5.4 ROCK Inhibitor Y-27632 Increases Fusion Efficiency of Droplet-Engineered Organoids

Although alginate reduced organoid differentiation and enhanced cell–cell and cell–matrix interactions, the extent of organoid fusion remained limited. Y-27632, a Rho-associated kinase (ROCK) inhibitor, has been reported to accelerate cellular fusion [155]. To evaluate its effect on DEO fusion, we cultured high-density DEOs with five concentrations of Y-27632 (0, 10, 20, 50, and 100 μM), maintaining the inhibitor in the culture medium during medium changes. Over a 7-day culture period, DEOs without Y-27632 (control) displayed close cell–cell contacts and organoid aggregation, with fusion occurring between days 3 and 5, but these organoids also exhibited earlier signs of differentiation compared to Y-27632-treated groups. Notably, addition of Y-27632 appeared to delay differentiation in a concentration-dependent manner, with 50 μM and 100 μM-treated DEOs maintaining more cystic, less differentiated morphologies compared to lower concentrations and controls (Figure 3.14).

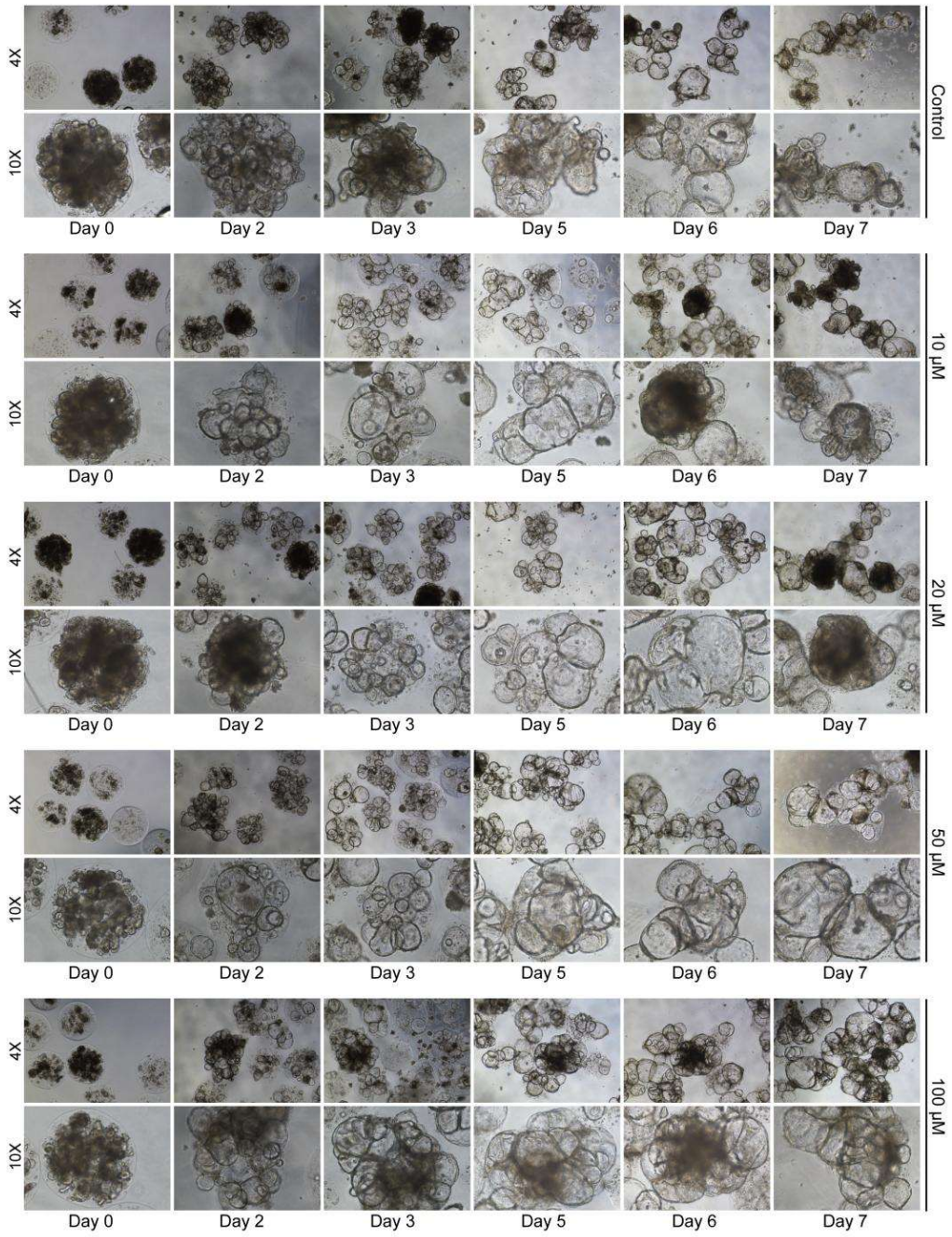


Figure 3.14 Y-27632 promotes cystic morphology and delays differentiation in DEOs. DEOs were cultured with increasing concentrations of the ROCK inhibitor Y-27632 (0, 10, 20, 50, and 100 μM) over 7 days. Brightfield images were captured to monitor organoid morphology. Images were acquired using 4× and 10× objective lenses, as indicated.

The addition of Y-27632 was found to promote organoid fusion, with the effect being particularly pronounced at 100 μM as early as day 3 (Figure 3.14). Representative 10× brightfield images captured on day 6 were used to compare fusion across experimental conditions. In the control group, limited fusion of organoid cells was observed. Increasing concentrations of Y-27632 corresponded with a progressive enhancement of fusion events, with both the frequency and number of fused organoids notably higher in the 50 μM and 100 μM treatment groups compared to lower concentrations and the control groups (Figure 3.15).

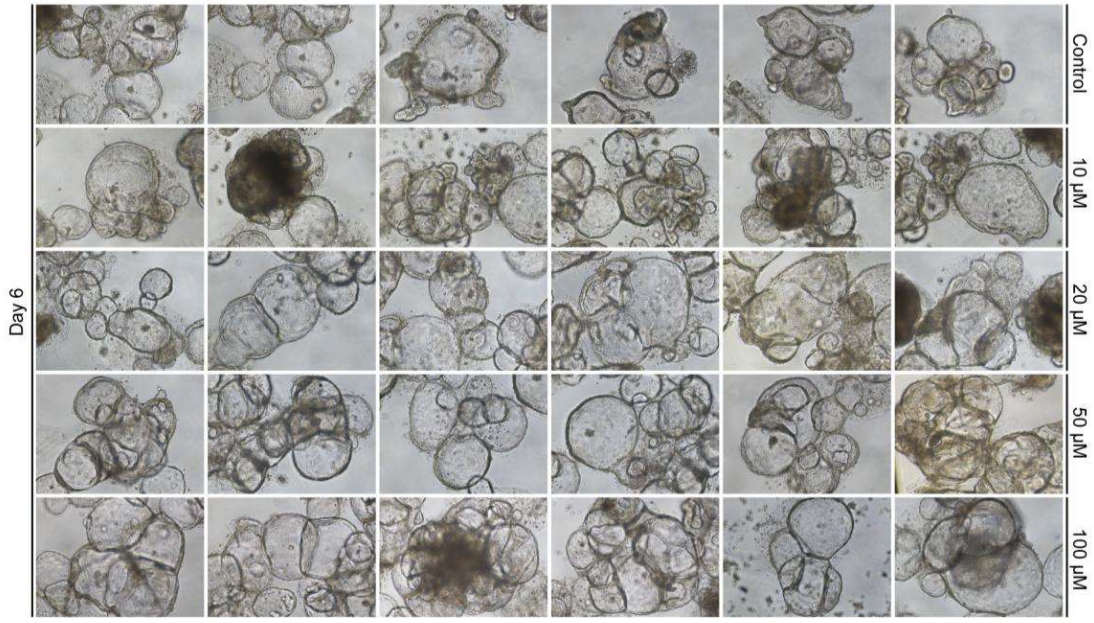


Figure 3.15 Effects of Y-27632 in DEOs. Representative 10× brightfield images of DEOs after 6 days of culture under increasing concentrations of Y-27632 (0, 10, 20, 50, and 100 μM).

3.6 Chapter Summary

In this chapter, robust and reproducible human intestinal organoid models were successfully established and systematically characterized, with a particular focus on appendix-derived gut organoids. Organoids were generated from multiple human intestinal tissues, including appendix, small intestine, and colon, and demonstrated typical epithelial cystic morphology, long-term expandability, and stability across serial